



CUSTOMER SUPPORT AGENT

Evaluation Report

A controlled evaluation journey from ground-truth review to prompt improvement, trace inspection, regression testing, and LLM-judge calibration.

Dataset	Model	Evaluation date
150 labeled tickets	Nebius gpt-oss-120b	September 6, 2026

WEEK 4 - EVALUATE YOUR AGENT

Executive summary

This project evaluated an e-commerce support classifier that routes each customer ticket to one of five queues. The work began with the instructor-provided 150-ticket workbook, reviewed 29 reference failures, identified recurring patterns, and selected Account_Workflow_Missed as the most defensible high-impact target. A controlled prompt experiment then improved the reproducible Nebius baseline from 94.0% to 99.3% accuracy.

94.0% Reproducible baseline	99.3% Improved accuracy	9 of 9 Target failures fixed	1 Regression
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Primary finding

The original account_help definition covered login, password, address, and payment-method changes, but omitted account-linked workflow failures. Adding one explicit rule corrected app crashes, rejected checkout promo codes, and purchases missing from account history.

Evaluation one-liner

I measured exact-match routing accuracy, account-workflow failure count, regression count, latency, and token usage on a fixed 150-ticket labeled dataset, using human-reviewed ground truth and LangSmith-traced baseline and improved runs. The pass bar was at least 95% accuracy, zero remaining target-category failures, no more than one regression, and p95 latency below 1.5 seconds.

The evaluation journey

1. Review Assess labels and isolate 29 supplied failures.	2. Explain Annotate each error with the competing cues.	3. Cluster Group recurring failures into three patterns.
4. Baseline Run the unchanged prompt on 150 fixed tickets.	5. Improve Add one focused account-workflow rule.	6. Validate Measure delta, regressions, traces, and judge agreement.

1. Evaluation design

Agent and user outcome

The agent predicts exactly one category: `order_status`, `refund_request`, `product_issue`, `account_help`, or `other`. The user outcome is accurate routing to the team that can resolve the primary problem without avoidable transfers or delays.

Fixed dataset

Category	Cases	Share	Examples of coverage
<code>order_status</code>	24	16%	Tracking, delivery status, non-delivery
<code>refund_request</code>	42	28%	Refunds, subscriptions, post-purchase adjustments
<code>product_issue</code>	24	16%	Damaged, wrong, defective, not as described
<code>account_help</code>	30	20%	Login, checkout, app access, order history
<code>other</code>	30	20%	General questions, feedback, product browsing
Total	150	100%	Happy paths, ambiguous cases, known failures

Ground-truth assessment

The supplied labels were treated as the assignment rubric and reviewed during failure annotation. Some refund-versus-underlying-problem cases require subjective tie-breaking and were flagged as higher-noise examples. The account-workflow cases formed a more internally consistent improvement target.

Metrics and pass bars

Metric	Judge method	Pass bar
Overall routing accuracy	Exact match	At least 95%
Account-workflow failures	Exact match + human grouping	0 after improvement
Regressions	Per-ticket before/after comparison	No more than 1
Latency	LangSmith trace duration	p95 below 1.5 s
Token usage	LangSmith token totals	Report as cost proxy

Trace design

LangSmith recorded one parent trace per ticket plus child spans for prompt construction, the Nebius model call, the runnable sequence, and structured-output parsing. Run name, prompt version, ticket ID, true category, prediction, and reasoning connect each aggregate score to an individual case. Correctness is stored in the paired result row by comparing true and predicted categories.

2. From reference failures to a reproducible baseline

The instructor reference run

The workbook supplied 121 correct predictions out of 150, or 80.7% accuracy, with 29 failures. We used those failures for qualitative analysis, not as the numerical baseline for our prompt experiment, because the original model configuration differs from the Nebius setup.

Failure category	Count	What the pattern captures	Why it matters
Refund_Intent_Missed	12	Money-back intent loses to delivery, policy, or account cues.	Financial requests reach the wrong team.
Account_Workflow_Missed	9	Checkout, app access, promo codes, or account history are routed by surface topic.	Customers can be blocked from paying or accessing purchases.
Scope_Boundary_Misread	8	Keywords override whether the case is shipment, purchased item, or general inquiry.	Specialist queues receive avoidable transfers.

Why Account_Workflow_Missed was selected

Refund_Intent_Missed was larger, but several examples depend on inconsistent tie-breaking between the underlying problem and the requested remedy. Account_Workflow_Missed contained nine consistently labeled, high-impact cases and offered a cleaner controlled test with less label noise.

Reproducible baseline

Measure	Result
Correct predictions	141 / 150
Accuracy	94.0%
p50 latency	0.603 s
p95 latency	0.936 s
Total tokens	45,145
Average tokens per ticket	301.0

Baseline failure pattern

All nine Nebius baseline failures were true account_help cases: app crashes while opening order history, rejected newsletter promo codes, and completed purchases missing from account history. This independently reinforced the target chosen from the instructor reference run.

3. One focused prompt experiment

Hypothesis

If the prompt explicitly defines account-linked workflow failures, the classifier should route the nine target cases to account_help without broadly over-routing unrelated order, product, refund, or general-inquiry cases.

The only classifier change

If the customer cannot complete an account-linked action because of repeated logout, an app or site crash, a rejected promo code at checkout, or a missing order record in account history, classify as account_help. Do not route only from surface words such as order, product, or discount.

Controlled variables

The same 150 tickets, Nebius gpt-oss-120b model, temperature 0, structured-output schema, evaluator logic, and LangSmith trace structure were retained. The rule above was the only substantive classifier change.

Measured result

Measure	Baseline	Improved	Delta
Correct predictions	141 / 150	149 / 150	+8
Accuracy	94.0%	99.3%	+5.3 points
Target failures	9	0	-9
Regressions	0	1	+1
p50 latency	0.603 s	0.574 s	-0.029 s
p95 latency	0.936 s	0.997 s	+0.061 s
Total tokens	45,145	56,161	+11,016
Average tokens	301.0	374.4	+73.4

Outcome

The prompt met every declared quality and latency pass bar. It fixed all nine targeted failures and increased accuracy by 5.3 percentage points. The tradeoff was a 24.4% increase in token usage and one regression.

4. LangSmith trace evidence - baseline t105

The original prompt classified the app crash as other. The trace shows the fixed ticket input, true category, incorrect output, model reasoning, original prompt version, child model call, latency, and token usage.

customer-support-baseline-ticket ID

Messages Beta Details

Trace

customer-support-baseline-ticket 1:21 PM

0.83s 320

RunnableSequence

0.83s 320

ChatPromptTemplate 0.00s

ChatOpenAI openai/gpt-oss-120b

0.83s 320

RunnableLambda 0.00s

customer-support-baseline-ticket ID

Feedback Input Output Attributes

Feedback

+ Add feedback

Input

JSON

```
{
  "ticket_id": "t105",
  "ticket_text": "Your app keeps crashing whenever I try to view my order history.",
  "true_category": "account_help"
}
```

RAW

Output

JSON

```
{
  "langsmith_run_id": "01a07862-7f92-7622-93a9-f042272b577e",
  "predicted_category": "other",
  "reasoning": "The issue is about the app crashing when accessing order history, which is a technical problem not covered by the defined categories."
}
```

RAW

Attributes

Metadata

evaluation_run	baseline
prompt_version	instructor-original-v1
revision_id	ef517b2
ticket_id	t105

4. LangSmith trace evidence - improved t105

The revised prompt classified the same ticket as `account_help`. The reasoning explicitly cites the new account-workflow rule, providing trace-level evidence that the intended intervention produced the correction.

customer-support-improved-ticket ID

Messages Beta Details

Trace

customer-support-improved-ticket 2:17 PM

0.80s 404

RunnableSequence

0.80s 404

customer-support-improved-ticket ID

Feedback Input Output Attributes

Input

```
{
  "ticket_id": "t105",
  "ticket_text": "Your app keeps crashing whenever I try to view my order history.",
  "true_category": "account_help"
}
```

RAW

Output

```
{
  "langsmith_run_id": "01a07895-78b9-7fb3-ba18-415f5867ff76",
  "predicted_category": "account_help",
  "reasoning": "The user cannot view order history due to app crashes, which is an account-linked action failing because of a technical issue. According to the disambiguation rule, this should be classified as account_help."
}
```

RAW

Attributes

Metadata

evaluation_run	improved
prompt_version	account-workflow-v2
revision_id	ef517b2
ticket_id	t105
true_category	account_help

LangSmith

5. Trace interpretation and LLM-as-a-judge

Case	Baseline	Improved	What the traces show
t105 - app crash	other	account_help	New rule applied to an account-history access failure.
t114 - promo code	other	account_help	Rejected checkout promo code moved into account_help.
t117 - missing history	order_status	account_help	Missing account record no longer treated as tracking.
t050 - price adjustment	refund_request	other	Single regression exposed a separate refund boundary gap.

Binary judge design

The optional judge evaluated all 29 instructor-reference failures. It saw only the customer message and the classifier's incorrect predicted intent. It returned TRUE when the error matched Account_Workflow_Missed and FALSE otherwise. Human Step 3 labels supplied the ground truth: nine TRUE and 20 FALSE cases.

Judge model	Agreement	p50	p95	Tokens	Decision
gpt-oss-120b	100% (29/29)	0.772 s	1.119 s	13,945	Selected
Qwen3-235B	100% (29/29)	0.927 s	1.283 s	10,046	Comparison

Judge selection

Both models achieved perfect precision, recall, and F1 for both classes. No prompt calibration was justified because there were no mismatches. gpt-oss-120b was selected for lower latency and more concise explanations; Qwen used fewer tokens but was slower and more verbose.

Representative judge traces

Case	Human label	Judge label	Trace finding
t103	TRUE	TRUE	Repeated logout during payment met both rubric conditions.
t024	FALSE	FALSE	Wrong delivery address correctly triggered the delivery exclusion.

6. What improved, what did not, and what comes next

The remaining regression: t050

Ticket: Item went on sale right after I ordered. Do you do price adjustments? The improved prompt changed the prediction from the correct refund_request route to other. The trace reasoning says that a post-purchase price adjustment is not covered by the defined categories.

Why we did not patch it in this experiment

Fixing t050 would require a second intervention: explicitly mapping post-purchase price adjustments to refund_request. Adding that rule now would weaken causal attribution because the reported improvement would combine two prompt changes. The regression is therefore reported honestly and reserved for a separate evaluation cycle.

Recommended next experiment

Step	Action	Regression check
1	Add one price-adjustment rule to refund_request.	Keep account-workflow v2 unchanged.
2	Rerun the same 150 tickets as prompt version v3.	Compare every case with v2.
3	Inspect discount-related cases and new flips.	Separate post-purchase adjustments from pre-purchase discounts.
4	Accept only if t050 is fixed without new regressions.	Retain LangSmith trace IDs and token/latency metrics.

Production monitoring

Retain prompt version, model, run name, ticket ID, expected category when available, predicted category, latency, tokens, and error status in LangSmith. Alert on labeled accuracy below 95%, reappearance of account-workflow failures, regression growth above 1%, p95 latency above 1.5 seconds, or token usage more than 25% above the improved baseline.

Final conclusion

Systematic evaluation changed the project from a plausible demo into a measurable improvement. Human review found the failure structure, a fixed dataset made the comparison fair, LangSmith connected aggregate metrics to individual decisions, regression testing exposed the remaining boundary gap, and the LLM-judge exercise demonstrated how an automated evaluator must be calibrated against human labels.